

Indian Institute of Technology Kharagpur

Department of Computer Science and Engineering

Class Test-1, Spring 2024-25

High Performance Computer Architecture (CS60003)

Students: 160

Date: 13-February-2025

Full marks: 25

Time: 10:00 AM – 10:55 AM

Credit: 10%

INSTRUCTIONS: This class-test is closed book and closed notes. Calculators are allowed. ANSWER ALL QUESTIONS.

1. In an assembly-language program P , 20% instructions require sequential execution because of data dependence. Two machines $M1$ and $M2$ are given: $M1$ comprises one core, i.e., a single processor; $M2$ is a multi-core machine with 80 copies of $M1$ on-chip that are capable of performing parallel processing. Both of them operate with the same clock frequency. Calculate the maximum achievable speed-up with respect to the execution time needed by the new machine $M2$ compared to that of $M1$ for the program P , following *Amdahl's Law*. [3]
2. Consider the following RISC-V code segment being executed: on a RISC-V microprocessor with a 5-stage integer pipeline unit:

```
lw    t1, 10(a0)
addi  t1, t1, 200
lui   t3, 0x7FFF
ori   t3, 0xFFFF
li    t2, 100 // initialize loop counter
Loop: xor  t4, t1, t3
      add  t1, t1, t4
      srl  t1, t1, 1
      addi t2, t2, -1 // decrement loop counter
      bne  t2, zero, Loop
```

Assume M has a clock frequency is 4.0 GHz, and each load/store/arithmetic/logic instruction takes 1 clock cycle to execute. Evaluation of the branch condition happens in the EX stage. Effective address calculation for load/store instructions take place in the ID state. All standard data forwarding arrangements are present. A “use after load” situation results in 1 clock cycle stall for the instruction immediately after a load instruction. The processor has an 1-bit branch predictor. The “bne” instruction incurs two clock cycles penalty if branch prediction is incorrect. The branch predictor is initialized to “Not Taken” state. Calculate the total number of clock cycles required to execute this code segment. [7]

3. Justify the following statement: “increase in TLB size does not help to reduce either cache miss rate or page-fault rate.” [3]
4. (a) Consider a 64-byte cache with 8 byte blocks, an associativity of 2 and LRU block replacement. Virtual addresses are 16 bits. The cache is physically tagged. The processor has 16KB of physical memory. What is the total number of tag bits? [2]
(b) Assume that the processor in part-(a) is part of a computer where each page is 64 bytes. How large would a single-level page table be given that each page requires 4 protection bits, and entries must be an integral number of bytes? [2]
5. Consider a CPU with only one level of cache memory.
 - (a) For a data cache with a 92% hit rate and a 2-cycle hit latency, calculate the *Average Memory Access Time* (AMAT). Assume that latency to memory and the cache miss penalty together is 124 cycles. Note: the cache must be accessed after memory returns the data. [2]

- (b) Calculate the performance of a processor taking into account stalls due to data cache and instruction cache misses. The data cache (for loads and stores) is the same as described in part-(a) above and 30% of instructions are loads and stores. The instruction cache has a hit rate of 90% with a miss penalty of 50 cycles. Assume the base CPI using a perfect memory system is 1.0. Calculate the CPI of the pipeline, assuming everything else is working perfectly. Assume the load never stalls a dependent instruction and assume the processor must wait for stores to finish when they miss the cache. Finally, assume that instruction cache misses and data cache misses never occur at the same time.
- i. Calculate the additional CPI due to the instruction cache stalls. [2]
 - ii. Calculate the additional CPI due to the data cache stalls. [2]
 - iii. Calculate the overall CPI for the machine. [2]